

Combinatorics – 6

Combinatorial geometry

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Theoretical minimum. Finitely many points in the plane, and a question about a configuration they must contain. The tools are the ones already proved elsewhere — pigeonhole, extremal choice, double counting — so this sheet also works as a map of where they live.

Definitions. A finite set of points is in *general position* if no three are collinear, and in *convex position* if all of them are vertices of their convex hull. For a set S of n points, $u(n)$ denotes the maximum number of pairs at distance exactly one, and $d(n)$ the minimum number of distinct distances.

Theorem 1 (Erdős–Szekeres, the happy ending). Among any five points in general position four are in convex position; and for every n there is $N(n)$ such that any $N(n)$ points in general position contain n in convex position. *Proof of the second part* from Ramsey’s theorem (Combinatorics – 5) by colouring quadruples, or from the monotone-subsequence theorem; the known bounds are $2^{n-2} + 1 \leq N(n) \leq \binom{2n-4}{n-2} + 1$, and the lower bound is conjectured to be the truth.

Theorem 2 (counting incidences). The number of unit distances satisfies $u(n) = O(n^{3/2})$: build the graph of unit-distance pairs; it contains no $K_{2,3}$, because two circles meet in at most two points, and a $K_{2,3}$ -free graph has $O(n^{3/2})$ edges by double counting the paths of length two. The same count gives $d(n) = \Omega(\sqrt{n})$. Both bounds are far from sharp: the truth for distinct distances is $n/\sqrt{\log n}$ by Guth and Katz, and for unit distances the answer is unknown.

Fact 1 (what stands elsewhere). Sylvester–Gallai — among finitely many points not all collinear there is a line through exactly two — is proved by the extremal principle in Methods of proof – 3. Helly’s and Radon’s theorems on

convex sets are Convex – 1. Minkowski’s lattice theorem and Pick’s formula are Convex – 10. The probabilistic bounds for colourings are Probability – 4. None of these is repeated here; each is used freely.

Fact 2 (colouring the plane). The chromatic number of the plane — the least number of colours needed so that points at distance one differ — satisfies $5 \leq \chi \leq 7$: the upper bound by a hexagonal tiling of diameter slightly less than one, the lower bound by explicit finite unit-distance graphs, the Moser spindle giving 4 and a computer-found graph giving 5.

1. Deduce the Erdős–Szekeres theorem on convex n -gons from Ramsey’s theorem, taking as known that among any five points in general position four are in convex position (that step is proved by the extremal principle in Methods of proof – 3, homework 2). Then prove the lower bound $N(n) \geq 2^{n-2} + 1$ by the standard construction, and explain why the Ramsey argument gives a bound doubly exponential in n .
2. Let S be a finite set of at least two points in the plane, no three collinear. A *windmill* starts with a line through one point $P \in S$ and rotates clockwise about P until it meets another point Q of S , which becomes the new pivot, and so on for ever. Show that one may choose the starting point and line so that the windmill uses each point of S as a pivot infinitely many times. (IMO shortlist 2011, C3)
3. Prove that the unit-distance graph on n points contains no $K_{2,3}$, deduce $u(n) = O(n^{3/2})$ by double counting the paths of length two, and deduce $d(n) = \Omega(\sqrt{n})$ for the number of distinct distances. Then show that $u(n) \geq n^{1+c/\log \log n}$ for the $\sqrt{n} \times \sqrt{n}$ grid.
4. Prove that $\mathbb{R}^2$ has a dense subset containing no three collinear points. (IMS 2007)
5. In a gathering of n people every two are friends or enemies, and m is the largest number of pairwise enemies. Represent every person by a point of $\mathbb{R}^2$ so that the distance between friends is at most 1 and between

enemies more than 3. Find the smallest k such that the people can always be split into k groups of pairwise friends. (IMS 2022)